Partial Likelihood and Cox Proportional Hazard Model

Module 17

Saurav De

Department of Statistics
Presidency University

Partial Likelihood and Cox Proportional Hazard Model

- Let T : lifetime of an item following continuous probability distribution and $P(T \geq 0) = 1$.
- $\overline{F}$(t) or S(t) or $P(T > t)$: Survival probability of the item at time point t.
- S(t) : known as survival function at time point t.
- H(t) = [- log_eS(t)] : cumulative hazard function at time point t.

Partial Likelihood and Cox Proportional Hazard Model

- $h(t) = \frac{d}{dt} H(t) = \frac{d}{dt}[-\ log_e S(t)]$: instantaneous failure/hazard rate at time point t.
- $h(t)$: known as hazard function at time point t.

  Note : $h(t) = \frac{-\frac{d}{dt}\overline{F}(t)}{\overline{F}(t)} = \frac{f(t)}{\overline{F}(t)}$, where $f(t)$: lifetime density at time point t.
- $S(t) \downarrow t \Leftrightarrow [-\ log_e S(t)] \uparrow t$. So, $h(t) = \frac{d}{dt}\ [-\ log_e\ S(t)] \geq 0, \forall\ t$.

Partial Likelihood and Cox Proportional Hazard Model

Survival Regression or Hazard Regression :

- Survival or hazard of an item at time t usually depends on some characteristics like gender, age, Body Mass Index (BMI), etc. Such characteristics are known as covariates.
- Let x denote a covariate supposed to affect $h(t)$, now expressed as $h(t, x)$, without loss of generality.
- Thus, survival or hazard regression can be looked upon as the functional dependence of $h(t, x)$ on covariate x.

Partial Likelihood and Cox Proportional Hazard Model

Purposes of hazard regression :

- To explore a mathematical model like $h(t, x) = g(x)$, where $g(x)$: suitable functional form.
- To predict an estimate of hazard corresponding to some covariate value.
- To measure effect of covariate x on hazard function, $h(t, x)$.
- To test the significance of covariate information x on $h(t, x)$.

Partial Likelihood and Cox Proportional Hazard Model

Cox's Proportional Hazard Model :

$$h(t, \mathbf{x}) = h_0(t, \boldsymbol{\alpha})\, e^{\boldsymbol{\beta}' \mathbf{x}}$$

, where $\mathbf{x}$ denotes a vector of covariates;
$\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ denote the parameters of the Cox's regression model.

Partial Likelihood and Cox Proportional Hazard Model

Some realisations on Cox's regression model :

- $h_0(t, \boldsymbol{\alpha})$: the value of $h(t, \mathbf{x})$ at $\mathbf{x} = \mathbf{0}$ (i.e., a baseline choice of $\mathbf{x}$). That is why, $h_0(t, \boldsymbol{\alpha})$ is called baseline hazard function at time point t.
- $h_0(t, \boldsymbol{\alpha})$ depends on t, but not on covariates.
- Term $e^{\boldsymbol{\beta}' \mathbf{x}}$ depends on covariates $\mathbf{x}$, but not on time point t.
- $\boldsymbol{\beta}$ measures the effects of $\mathbf{x}$ on $h(t, \mathbf{x})$. How ?
  For univariate $\mathbf{x}$, denoted by x, $\frac{h(t,x+1)}{h(t,x)} = e^{\beta} \Rightarrow \beta$ gauges the impact of change in $h(t, x)$ for unit change in x value.

Partial Likelihood and Cox Proportional Hazard Model

Some realisations on Cox's regression model (continued) :

- Consider two individuals with covariates $\mathbf{x_1}$ and $\mathbf{x_2}$. Then, the ratio of their hazards at time t is given by :

$$\frac{h(t, \mathbf{x_1})}{h(t, \mathbf{x_2})} = \frac{h_0(t, \boldsymbol{\alpha})\, e^{\boldsymbol{\beta}'\, \mathbf{x}_1}}{h_0(t, \boldsymbol{\alpha})\, e^{\boldsymbol{\beta}'\, \mathbf{x}_2}} = e^{\boldsymbol{\beta}'\, (\mathbf{x}_1 - \mathbf{x}_2)},$$

which is constant with respect to time t

Partial Likelihood and Cox Proportional Hazard Model

$\Rightarrow h(t, \mathbf{x_1}) \propto h(t, \mathbf{x_2})$ with respect to time t.

- In other words, the hazards are proportional $\forall\, t$. Hence, the name *Proportional Hazard Model (P.H.M.)*.

  i.e. hazard ratio of two individuals at age 8 years and age 80 years will remain same. It seems to be little unrealistic at times.

Partial Likelihood and Cox Proportional Hazard Model

Some realisations on Cox's regression model (continued) :

- In Cox's P.H.M., one will be interested in parameters $\boldsymbol{\beta}$, which measure the effects of covariates on survival, but usually we are not interested in $\boldsymbol{\alpha}$.
  $\Rightarrow$ $\boldsymbol{\beta} \rightarrow$ parameters of interest, $\boldsymbol{\alpha} \rightarrow$ nuisance parameters.
- Hence, no form is pre-specified for baseline hazard, $h_0(t, \boldsymbol{\alpha})$. The Cox's P.H.M. is thus called a semi-parametric model.

Partial Likelihood and Cox Proportional Hazard Model

Estimating the covariate parameters β : Partial Likelihood

- As here, the form of $h_0(t, \alpha)$ is not specified, so the form of likelihood function is not known properly $\Rightarrow$ the usual method of maximum likelihood fails.
- For estimating the regression parameters β, Cox developed a non-parametric method and called it as *Partial Likelihood*.

Partial Likelihood and Cox Proportional Hazard Model

- Let m = number of individuals under study.
  $\delta_i = 1$, if i^{th} individual is uncensored.
  $= 0$, if i^{th} individual is right-censored. (i = 1,. . . , m).
- Define R(t_i) = set of all individuals surviving or functioning at time t_i $\rightarrow$ Risk set at t_i.
- Let $h_j(t)$ denote hazard of jth individual at time t.
- Then, given that t_i is an event time (i.e., failure/death time), the probability that individual i has that event is given by :

$$P([i]|t_i) = \frac{h_i(t_i)}{\sum_{j \in R(t_i)} h_j(t_i)}, \; [i] : individual\ i.$$

Partial Likelihood and Cox Proportional Hazard Model

- As per Cox's P.H.M. assumptions, $h_j(t, \mathbf{x_j}) = h_0(t, \boldsymbol{\alpha})\, e^{\boldsymbol{\beta}' \mathbf{x_j}}$

  $$\Rightarrow P([i]|t_i) = \frac{h_0(t_i, \boldsymbol{\alpha})\, e^{\boldsymbol{\beta}' \mathbf{x_i}}}{\sum_{j \in R(t_i)} h_0(t_i, \boldsymbol{\alpha})\, e^{\boldsymbol{\beta}' \mathbf{x_j}}} = \frac{e^{\boldsymbol{\beta}' \mathbf{x_i}}}{\sum_{j \in R(t_i)} e^{\boldsymbol{\beta}' \mathbf{x_j}}} = \frac{\phi_i}{\sum_{j \in R(t_i)} \phi_j}, (say)$$

  where, $\phi_j = e^{\boldsymbol{\beta}' \mathbf{x_j}}$.
- $P([i]|t_i)$ may be called as Risk Probability of individual i at time point t_i.

Partial Likelihood and Cox Proportional Hazard Model

Partial Likelihood for unique failure times :

- Suppose at each event time only one individual observes the event. Also, the individual observes the events independently. $\Rightarrow$ The partial likelihood for $\boldsymbol{\beta}$ is given by :
  $L_p(\boldsymbol{\beta}, x) = \prod_{i=1}^{m} \left[\frac{\phi_i}{\sum_{j \in R(t_i)} \phi_j} \right]^{\delta_i} = \prod_{i=1}^{m} \left\{ P([i]|t_i) \right\}^{\delta_i}.$
- Power δ_i means the individuals having the events (failure/death) actually contribute to the likelihood, but not the right-censored cases.

Partial Likelihood and Cox Proportional Hazard Model

Why the name *partial* ? Two reasons for that are :

1. L_p is not the full likelihood function of both the parameters α and β, but β only.
2. L_p does not use the full data; as not the actual times of occurrences are important, but their rankings are, e.g., the likelihood remains same no matter if the individuals i, j and k have event times $\{1,2,3\}$ or $\{10,80,94\}$.

Merit of partial likelihood method : The method involves less assumptions and hence is more robust than full likelihood method.
Demerit of partial likelihood method : The method is less powerful compared to a fully parametric model.

Partial Likelihood and Cox Proportional Hazard Model

Note : Partial likelihood acts exactly in a similar way as the full likelihood.

Let $\widehat{\boldsymbol{\beta_p}}$ denote the *Maximum Partial Likelihood* estimate of $\boldsymbol{\beta}$.

$$\Rightarrow L_p(\widehat{\boldsymbol{\beta_p}}) = \sup_{\boldsymbol{\beta_p}} \prod_{i=1}^{m} \left[\frac{e^{\boldsymbol{\beta}' \mathbf{x_i}}}{\sum_{j \in R(t_i)} e^{\boldsymbol{\beta}' \mathbf{x_j}}} \right]^{\delta_i}.$$

Partial Likelihood and Cox Proportional Hazard Model

Properties of $\widehat{\beta_p}$: Maximum Partial Likelihood estimate of β

1. $\widehat{\beta_p} \xrightarrow{P} \beta$ as m $\to \infty$.
2. $Disp(\widehat{\beta_p}) \approx I_p^{-1}$, where I_p is calculated from L_p exactly in the same way as usual *Information Matrix* from full likelihood.
3. $\widehat{\beta_p} \xrightarrow{\mathscr{D}} \mathrm{N}(\beta, I_p^{-1})$ as m $\to \infty$.

Partial Likelihood and Cox Proportional Hazard Model

Partial Likelihood for repeated failure times :

The case when two or more individuals may have same event times. Let $t_{(i)}$ be i^{th} distinct ordered event time, i.e., $t_{(i)} < t_{(i+1)}$ (that means, ordered) and four failure times 1, 1, 3, 3 $\Rightarrow t_{(1)} = 1$ and $t_{(2)} = 3$ (that means, distinct).

- I = number of unique event times, and
- D(t) be the set of individuals having event at time point t.

Partial Likelihood and Cox Proportional Hazard Model

- Under repeated failure times, there are three popular methods of framing likelihood, which are :
  1. Breslow's method.
  2. Efron's method.
  3. Exact method.
- Consider the following situation first.
  Suppose individuals labelled 1-5 are at risk of failing at $t_{(i)} \Rightarrow R(t_{(i)}) = \{ 1, 2, 3, 4, 5 \}$.
- Let individuals 1-3 actually fail at $t_{(i)} \Rightarrow D(t_{(i)}) = \{ 1, 2, 3 \}$.

Partial Likelihood and Cox Proportional Hazard Model

Then Breslow's method contributed from time $t_{(i)}$ to L_p by :

$\frac{\phi_1}{(\phi_1+\cdots+\phi_5)} \times \frac{\phi_2}{(\phi_1+\cdots+\phi_5)} \times \frac{\phi_3}{(\phi_1+\cdots+\phi_5)}$

$\Rightarrow$ formally, we have from <u>Breslow's method</u> that :

$$L_p(\beta, x) = \prod_{i=1}^{I} \left\{ \prod_{j \in D(t_i)} \frac{\phi_j}{\left(\sum_{j \in R(t_j)} \phi_j\right)} \right\} = \prod_{i=1}^{I} \left\{ \frac{\prod_{j \in D(t_{(i)})} \phi_j}{\left(\sum_{j \in R(t_{(i)})} \phi_j\right) \mid D(t_{(i)}) \mid} \right\}$$

where, $\mid D(t_{(i)}) \mid =$ number of individuals in set $D(t_{(i)})$.

It assumes that all the 3 individuals fail simultaneously at $t_{(i)} \rightarrow$ a crude and unrealistic assumption.

Partial Likelihood and Cox Proportional Hazard Model

Due to Efron's method contribution to L_p through illustration is :

$$\frac{\phi_1\,\phi_2\,\phi_3}{\left(\phi_1+\cdots+\phi_5\right)\left(\phi_1+\cdots+\phi_5-\frac{1}{3}(\phi_1+\phi_2+\phi_3)\right)\left(\phi_1+\cdots+\phi_5-\frac{2}{3}(\phi_1+\phi_2+\phi_3)\right)}.$$

It assumes non-instanteinity in failure of 3 individuals and accordingly denominators are adjusted.
As it is not known a priori that who is the first to fail, so one-third of $(\phi_1+\phi_2+\phi_3)$ is adjusted from $\sum_{j=1}^{5}\phi_j$ after one fails. Similarly two-third of $(\phi_1+\phi_2+\phi_3)$ is adjusted after first two individuals fail.

Partial Likelihood and Cox Proportional Hazard Model

So, in general, under <u>Efron's method</u> :

$$L_p(\boldsymbol{\beta}, x) = \prod_{i=1}^{l} \frac{\prod_{j \in D(t_{(i)})} \phi_j}{\prod_{k=1}^{|D(t_{(i)})|} \left[\sum_{j \in R(t_{(i)})} \phi_j - \frac{k-1}{|D(t_{(i)})|} \sum_{j \in D(t_{(i)})} \phi_j \right]}.$$

This method is much more accurate. Also this method is the default method of partial likelihood for fitting Cox PHM using in R saftware.

Partial Likelihood and Cox Proportional Hazard Model

Using the illustration, finally Exact method contributes :

$$\frac{\phi_1\phi_2\phi_3}{\phi_1\phi_2\phi_3 + \phi_1\phi_2\phi_4 + \phi_1\phi_2\phi_5 + \cdots + \phi_3\phi_4\phi_5}.$$

$\Rightarrow$ in general, <u>Exact method</u> defines :

$$L_p(\boldsymbol{\beta}, x) = \prod_{i=1}^{I} \frac{\sum_{j \in D(t_{(i)})} \phi_j}{\sum_{j \in Q_i} \Phi_q}.$$

where, Q_i = set of all $\mid D(t_{(i)}) \mid$-tuples that could be selected from $R(t_{(i)})$ and Φ_q = the product of ϕ_j's over all members j of a $\mid D(t_{(i)}) \mid$-tuple q.

Note :

- Efron's method is closer to Exact method.
- Under untied case, these three methods are just same with L_p for unique failure time.

Partial likelihood estimate by numerical method:

The above three methods of forming partial likelihood function are quite difficult for the purpose of fitting the (Cox PH) model parameters analytically.
Let $\boldsymbol{\beta}$ be $k-$dimensional model parameter vector and our objective is to get $\hat{\boldsymbol{\beta}}$ that maximises the log partial likelihood function $\ell_p(\boldsymbol{\beta})$.
Newton-Raphson method, as discussed below, can be used to get the estimate of parameters :

- Choose an arbitrary value $\boldsymbol{\beta}^{(0)}$ as an initial approximation to $\boldsymbol{\beta}$.

Partial Likelihood and Cox Proportional Hazard Model

- Define score vector

$$U(\boldsymbol{\beta}) = \left(\frac{\partial \ell_p(\boldsymbol{\beta})}{\partial \beta_1}, \ldots, \frac{\partial \ell_p(\boldsymbol{\beta})}{\partial \beta_k} \right)'$$

  and the $k \times k$ information matrix $I(\boldsymbol{\beta})$ whose (i,j)th element is $\frac{\partial^2 \ell_p(\boldsymbol{\beta})}{\partial \beta_i \partial \beta_j}$.

- Get the first approx. $\boldsymbol{\beta}^{(1)} = \boldsymbol{\beta}^{(0)} + I^{-1}(\boldsymbol{\beta}^{(0)}) U(\boldsymbol{\beta}^{(0)})$

  Second approx. $\boldsymbol{\beta}^{(2)} = \boldsymbol{\beta}^{(1)} + I^{-1}(\boldsymbol{\beta}^{(1)}) U(\boldsymbol{\beta}^{(1)})$ and so on.

- The iterative method will converge at $r+1$th step if $\boldsymbol{\beta}^{(r)}$ and $\boldsymbol{\beta}^{(r+1)}$ agree upto certain decimal places, and then ml estimate $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}^{(r)}$ or $\boldsymbol{\beta}^{(r+1)}$.

- Further $\boldsymbol{\beta}^{(0)}$ from $\hat{\boldsymbol{\beta}}$, more is r i.e. less likely is the convergence to $\hat{\boldsymbol{\beta}}$.

Partial Likelihood and Cox Proportional Hazard Model

An Illustration. Let x_i : the body mass index (BMI) of individual i. Suppose a study is conducted on individuals who suffered from heart attack. the following 9 individuals died and t_i : the day(s) to death for the individual i following the attack. Data are:

i	1	2	3	4	5	6	7	8	9
t_i	6	98	189	374	1002	1205	2065	2201	2421
x_i	31.4	21.5	27.1	22.7	35.7	30.7	26.5	28.3	27.9

Partial Likelihood and Cox Proportional Hazard Model

Here Cox PHM : $h(t_i, x_i) = h(t_i, \boldsymbol{\alpha})e^{\beta x_i}$.
The partial loglikelihood will be

$$\ell_p(\beta) = \beta \sum_{i=1}^{9} x_i - \sum_{i=1}^{9} \log \left(\sum_{j \in R(t_i)} e^{\beta x_i} \right)$$

$$\implies \frac{\partial}{\partial \beta} \ell_p(\beta) = \sum_{i=1}^{9} x_i - \sum_{i=1}^{9} \frac{\sum_{j \in R(t_i)} x_j e^{\beta x_i}}{\sum_{j \in R(t_i)} e^{\beta x_i}} \quad \ldots (*)$$

Partial Likelihood and Cox Proportional Hazard Model

Hence

$$\frac{\partial^2}{\partial\beta^2}\ell_p(\beta) = -\sum_{i=1}^{9}\frac{A}{\left(\sum_{j\in R(t_i)} e^{\beta x_i}\right)^2}, \ (**)$$

where $A = \left(\sum_{j\in R(t_i)} x_j^2 e^{\beta x_i}\right)\left(\sum_{j\in R(t_i)} e^{\beta x_i}\right) - \left(\sum_{j\in R(t_i)} x_j e^{\beta x_i}\right)^2$.

Obviously here $U(\beta) = \frac{\partial}{\partial\beta}\ell_p(\beta)$ and $I(\beta) = -\frac{\partial^2}{\partial\beta^2}\ell_p(\beta)$.

$\Longrightarrow$ the Newton-Raphson recursive update relation is

$$\beta^{(k+1)} = \beta^{(k)} + \frac{U(\beta^{(k)})}{I(\beta^{(k)})}, \ k = 0, 1, \ldots$$

Partial Likelihood and Cox Proportional Hazard Model

Let us choose $\beta^{(0)} = 0$.
Hence from $(*)$ and $(**)$ we compute $U(0) = -2.512$ and $I(0) = 77.13$.
So

$$\beta^{(1)} = 0 - 2.512/77.13 = -0.0326$$

This way we get (on computation)

$$\beta^{(2)} = -0.0326 - 0.069/72.83 = -0.0335 \ ,$$

$$\beta^{(3)} = -0.0335 - 0.000061/72.70 \approx -0.0335$$

As both $\beta^{(2)}$&$\beta^{(3)}$ agree upto 4 decimal places $\Longrightarrow$ the iteration converges correct upto 4 decimal places.
Hence, the ML estimate $\hat{\beta} = -0.0335$(correct upto 4 decimal places).

Fitting Cox PHM with partial likelihood method using R :-

Consider the similar type of data as follows:

i	1	2	3	4	5	6	7	8	9	10
t_i	6	98	98	98	98	189	374	374	374	1002
δ_i	1	1	1	0	1	1	0	1	1	1
x_i	31.4	21.5	23.2	22.5	25	27.1	22.7	22.7	20.6	35.7
z_i	89	93	69	58	84	81	81	74	79	70

Contd ...

Partial Likelihood and Cox Proportional Hazard Model

i	11	12	13	14	15	16	17	18
t_i	1002	1205	1205	1205	2065	2201	2201	2421
δ_i	1	1	0	1	1	1	0	1
x_i	24.7	30.7	25.1	28.8	26.5	28.3	29.5	27.9
z_i	76	73	67	70	86	71	75	73

where t_i : days to death or right censoring for ith individual,
δ_i : Censorship indicator for individual i; $\delta_i = 1$, if died; $\delta_i = 0$, if censored,
(x_i, z_i) : (BMI , Age) for individual i.

Using R, fit Cox PHM and study different inferential aspects of your fitting based on these repeated-event-time data.

Partial Likelihood and Cox Proportional Hazard Model

Fitting through Breslow's, Efron's and Exact Method of partial likelihood using R:

R Code and Output :

```
> library(survival)
> t=c
    (6,98,98,98,98,189,374,374,374,1002,1002,1205,1205,1205,2065,2201

> delta=c(1,1,1,0,1,1,0,1,1,1,1,1,0,1,1,1,0,1)
> bmi=c
    (31.4,21.5,23.2,22.5,25,27.1,22.7,22.7,20.6,35.7,24.7,30.7,25.1,28

> age=c(89,93,69,58,84,81,81,74,79,70,76,73,67,70,86,71,75,73)
> df=data.frame(t,delta,bmi,age)
> a=coxph(Surv(df$t, df$delta) ~ df$bmi + df$age,df,method = "
    breslow")
```

Partial Likelihood and Cox Proportional Hazard Model

R Code and Output (continued) :

```
> summary(a)
Call:
coxph(formula = Surv(df$t, df$delta) ~ df$bmi + df$age, data = df,
    method = "breslow")

  n= 18, number of events= 14

           coef exp(coef) se(coef)      z Pr(>|z|)
df$bmi -0.01820   0.98197  0.10412 -0.175   0.8613
df$age  0.07958   1.08283  0.04461  1.784   0.0745 .
---
Signif. codes:  0    ***    0.001    **    0.01    *    0.05    .
        0.1          1
```

Partial Likelihood and Cox Proportional Hazard Model

R Code and Output (continued) :

```
        exp(coef) exp(-coef) lower .95 upper .95
df$bmi      0.982     1.0184    0.8007     1.204
df$age      1.083     0.9235    0.9922     1.182

Concordance= 0.708  (se = 0.105 )
Rsquare= 0.196   (max possible= 0.963 )
Likelihood ratio test= 3.92  on 2 df,   p=0.1407
Wald test            = 4.1  on 2 df,   p=0.1284
Score (logrank) test = 4.31  on 2 df,   p=0.1158
> b=coxph(Surv(df$t, df$delta) ~ df$bmi + df$age,df,method = "efron
    ")
> summary(b)
Call:
coxph(formula = Surv(df$t, df$delta) ~ df$bmi + df$age, data = df,
    method = "efron")

  n= 18, number of events= 14
```

Partial Likelihood and Cox Proportional Hazard Model

R Code and Output (continued) :

```
            coef  exp(coef)  se(coef)       z  Pr(>|z|)
df$bmi  -0.02690    0.97346   0.10567  -0.255    0.7991
df$age   0.08192    1.08537   0.04461   1.836    0.0663 .
---
Signif. codes:  0    ***    0.001    **    0.01    *    0.05    .
        0.1          1

       exp(coef) exp(-coef) lower .95 upper .95
df$bmi    0.9735     1.0273    0.7913     1.197
df$age    1.0854     0.9213    0.9945     1.185

Concordance= 0.692  (se = 0.105 )
Rsquare= 0.209   (max possible= 0.961 )
Likelihood ratio test= 4.22  on 2 df,   p=0.1211
Wald test            = 4.42  on 2 df,   p=0.11
Score (logrank) test = 4.66  on 2 df,   p=0.0975
```

Partial Likelihood and Cox Proportional Hazard Model

R Code and Output (continued) :

```
> d=coxph(Surv(df$t, df$delta) ~ df$bmi + df$age,df,method = "exact
    ")
> summary(d)
Call:
coxph(formula = Surv(df$t, df$delta) ~ df$bmi + df$age, data = df,
    method = "exact")

  n= 18, number of events= 14

            coef exp(coef) se(coef)      z Pr(>|z|)
df$bmi -0.02102   0.97920  0.10851 -0.194   0.8464
df$age  0.09451   1.09912  0.05071  1.864   0.0624 .
---
Signif. codes:  0    ***    0.001    **    0.01    *    0.05    .
        0.1         1
```

Partial Likelihood and Cox Proportional Hazard Model

R Code and Output (continued) :

```
        exp(coef) exp(-coef) lower .95 upper .95
df$bmi     0.9792     1.0212    0.7916     1.211
df$age     1.0991     0.9098    0.9951     1.214

Rsquare= 0.221   (max possible= 0.94 )
Likelihood ratio test= 4.5  on 2 df,   p=0.1052
Wald test            = 4.3  on 2 df,   p=0.1162
Score (logrank) test = 4.75  on 2 df,   p=0.09296
```